



Editorial

Are we finally tackling the issue of pain during cesarean section?



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There has been much emphasis and numerous publications in the last decade on strategies to optimize pain management *after* cesarean section, and stepwise opioid-sparing multimodal analgesia within an enhanced recovery pathway is recognized as the ultimate goal to achieve [1,2]. Indeed, women experiencing pain after cesarean section have been shown to be at higher risk for persistent pain, postpartum depression and even new onset opioid use dependence [3,4].

There has been much less focus on how to prevent and manage pain that might occur *during* cesarean section, even though this undoubtedly causes unacceptable suffering and a potentially devastating experience for the mother and her support system. When asked about their preferences and concerns, pregnant women considered pain during and after cesarean delivery to be their most important concern, unlike nausea and vomiting, which general surgical patients rate the highest [5]. Pain during cesarean section is also a cause for litigation [6].

There are multiple reasons for a patient to experience discomfort or pain during cesarean section, some of which are preventable, and all should certainly be managed appropriately. Pain during cesarean section under neuraxial anesthesia represents a challenging clinical situation, where the anesthesiologist(s) will need to make swift decisions based on the obstetrical (emergent *versus* not urgent delivery) and surgical (before or after skin incision) contexts along with patient-specific characteristics (co-morbidities, anxiety, preferences). Based on those circumstances, options might include an additional anesthetic procedure, such as repeating the neuraxial procedure or converting to general anesthesia; both are associated with risks and complications. Repeating a neuraxial procedure has been associated with high neuraxial block with potentially devastating complications [7]. The role of general anesthesia for cesarean section has recently been revisited [8] and thoughts are that it should be avoided whenever possible [9,10]. Therefore, general

anesthesia rates for cesarean section serve as an outcome metric for quality of obstetric anesthesia care; it is now agreed that the general anesthesia rate for cesarean section should be around 5–10% based on circumstances [11]. Consequently, coupled with challenges attributed to the “pregnant airway” [12], it is expected that anesthesiologists will consider all possible alternatives before converting to general anesthesia. The question now is how to mitigate the risk of discomfort and pain during cesarean section *versus* that associated with repeating a neuraxial procedure (if amenable) or converting to general anesthesia?

We are gratified to see The French Practice Bulletin published in this issue of *Anaesthesia Critical Care & Pain Medicine*, which is adapted from the French “Preconisations”, written with the goal to raise global awareness about the issue of acute pain during cesarean section and provide a framework for healthcare providers caring for pregnant women on how to prevent, recognize and manage failed or insufficient neuraxial anesthesia for cesarean section. The authors should be congratulated for tackling the question of pain during cesarean section: to our knowledge, (sub)specialty organizations have remained silent on this prickly topic.

The authors describe the process by which an interdisciplinary multi-professional taskforce was convened to improve patient-centered care during cesarean section. The taskforce identified 5 themes and developed 23 statements of which 10 were deemed key to decrease the likelihood of intraoperative pain and post-traumatic stress. These 5 themes focus on the (1) *prevention* of failed/insufficient anesthesia describing what constitutes appropriate neuraxial anesthesia and testing of the surgical block prior to incision, (2) *communication* surrounding the appropriate decision to delivery time (3) *recognition* and *response* to pain reported before and/or after skin incision, (4) *screening, prevention and management* of post-traumatic stress disorder, and (5) *mitigation and litigation*.

First, they provided a novel definition for what constitutes an insufficient/failed neuraxial anesthesia which they defined as “any circumstance when, under neuraxial anesthesia, the patient requests additional medication”. Essential to this definition is that anxiety and discomfort expressed by the patient must be taken into consideration and are as important as the “objective” testing of the surgical block. In other words, it is important to listen to, and believe the patient when she complains of pain and it is equally important to take action [13]; telling the patient that the anesthetic is working, and that what she is feeling is “just pressure and is normal” is not good enough. Poor communication between women and their anesthesiologists has resulted in complaints and litigation, with many lessons learned from reviewing such cases [6,14,15].

Further, they describe how to prevent failures by checking the adequacy of a neuraxial block for cesarean section, which should be assessed by both the anesthesiologist (symmetrical block to light touch at T6 ± to cold at T3) and the obstetrician (skin pinch test at least at the umbilicus prior to incision). A list of patient-specific (e.g. high body mass index and spine surgery), procedure specific (e.g., neuraxial procedure performed at a low level (L5-S1) or partial injection of spinal dose, intrapartum breakthrough pain with more than two epidural boluses and elevated pain scores within 2 h prior of cesarean section) and obstetric-specific factors (e.g., urgent cesarean section) associated with failed/insufficient neuraxial anesthesia for cesarean section is provided. Of note, additional patient-specific factors such as previous history of pain during cesarean section, fear of pain, chronic pain or opioid-use disorder are not listed, which could guide neuraxial dosing in a tailored manner. The authors provide a decision tree that proposes several rescue strategies ranging from how to proceed with a second neuraxial anesthetic, to administering intravenous supplementation or inducing general anesthesia.

The importance of continued communication between the obstetrical and anesthesia teams is highlighted, particularly surrounding suspected fetal acidosis and the need to urgently proceed with delivery, without omitting to monitor again the fetal heart rate upon arrival to the operating room, as this will provide guidance as to the emergent nature of the delivery. The intraoperative huddle between obstetricians and anesthesiologists is also key and will allow teams to decide together of next steps, based on obstetrical circumstances. At all times should the decision to proceed (or not) with a general anesthetic be considered. Documentation and debriefing of all cases with involved teams allows identifying opportunities for improvement, and auditing is highly encouraged. Last, a follow-up with postpartum mental health specialists is necessary to ensure that patients express any traumatic experience and are able to receive the appropriate care in the setting of post-traumatic stress. Ultimately, chronological steps proposed in the French Practice Bulletin may be summarized as follows:

- (1) Communicate with the patient and set expectations about (dis)comfort during cesarean section, describe possible rescue modalities if the neuraxial block is providing sub-optimal anesthesia and obtain informed consent.
- (2) Check appropriately the surgical block before skin incision and be ready to take action if needed (rescue plan may include repeating the neuraxial block if time allows, supplementing intravenously or *via* epidural catheter, or inducing general anesthesia).
- (3) Believe your patient if she tells you the block is insufficient, or if the surgery has started and she is uncomfortable during the cesarean section.
- (4) Communicate with the obstetricians and discuss the above anesthetic rescue options based on urgency to deliver the baby and whether the surgery has already started.

- (5) Call for an intraoperative huddle with the obstetricians if pain during cesarean section occurs, and based on circumstances, supplement or induce general anesthesia.
- (6) Provide rescue strategy based on patient specific, obstetric and surgical circumstances.
- (7) Document, debrief and audit all cases with the teams to allow quality improvement processes.
- (8) Follow-up all patients who experienced intraoperative pain to identify and manage psychological harm that may result in post-traumatic stress disorder.

While the framework is extremely useful, The French Practice Bulletin suffers several shortcomings. First, it does not offer a systematic review of the literature of publications reporting on cases of failed/insufficient anesthesia with pain during cesarean section (7 articles are cited) nor does it provide an exhaustive review of publications on failed spinal anesthesia (1 review article is cited) or intrapartum epidural failure conversions (6 papers are cited). Second, while the process by which statements or expert opinions were made did follow an iterative approach to reach agreement between 30 members of the taskforce, recommendations were not categorized by class of recommendation and level of evidence according to the American College of Cardiology/American Heart Association (ACC/AHA) classification system, as usually applied to clinical practice guideline recommendations [16]. Third, as proposed in two recent consensus statements published by the Society for Obstetric Anesthesia and Perinatology (SOAP) [17,18] that provide guidance in challenging situations where anesthesiologists need to mitigate risks of neuraxial procedures *versus* risks associated with alternatives (i.e., general anesthesia for urgent cesarean section), a risk assessment is not proposed here. Here, the risks associated with general anesthesia *versus* those of long-term psychological trauma need to be weighed, although inducing general anesthesia without other actions may not suffice to ensure psychological safety [14]. Fourth, the clinical decision tree proposes some strategies to rescue insufficient/failed anesthesia before skin incision or during cesarean section, but they are not extensive, and the anesthesiologist encountering such situation may want to include additional pharmacological options; neither alpha-2 agonists (clonidine or dexmedetomidine) that may be effectively added neuraxially or intravenously, nor anxiolytics (e.g., benzodiazepines) are suggested. It may well be that specific pharmacological recommendations will need to be adapted to institutional practices (e.g., alfentanil and remifentanil may be the short-acting opioids of preference in France, but that is not the case everywhere). Last, while the goal of a patient-centric approach to prevent unnecessary suffering and a traumatic birthing experience is emphasized throughout the practice bulletin (i.e., informing, communicating, consenting for all possible procedures and acknowledging discomfort), shared-decision making is not mentioned. In this context, discussing patient's preference is helpful and might allow gauging the amount of pain and discomfort experienced. In circumstances where the partner is present in the operating room, engaging the patient and the partner in the decision to induce general anesthesia may not only increase trust and communication, it will also ensure some equality in the doctor-patient relationship and empower the patient and partner.

Ultimately, listening to the patient, before, during and after the cesarean section is key. While the failure of “our” neuraxial anesthetic may be difficult to accept, our true failure is when we fail to acknowledge that the patient is experiencing insufferable discomfort or pain and we do not act upon it appropriately. It has been said that the “golden rule” is to treat our patients the way we would like to be treated, however, the “platinum rule” is to treat them the way they want to be treated.

Conflicts of interest

The authors declare no conflict of interest

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Ruth Landau*, Anne-Sophie Janvier

Department of Anesthesiology, Columbia University, 622 West 168th St,
PH-5, New York, NY 10032, USA

*Corresponding author

E-mail address: rl262@cumc.columbia.edu (R. Landau).

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